The chain

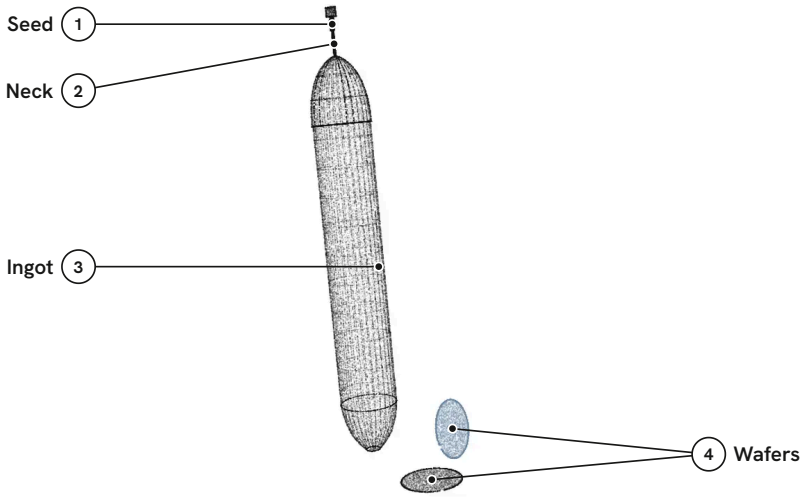
01 Chip Design, EDA and IP
02 Silicon and Wafers
03 Chemicals, Gases and Photoresist
04 Lithography
05 Photomasks and Pellicles
06 Deposition and Etch
07 Metrology and Inspection
08 Foundries and Fabs
09 Memory and HBM
10 Advanced Packaging
11 Substrates and PCBs
12 Test and Assembly
13 Systems and Networking
14 Data Centers and Power

15 The Economics of AI Chips
16 Who Controls the Chain

Stage 2 of 14 · High concentration · Moderately hard to substitute

Silicon and Wafers

Firms based in Japan, Taiwan, Germany and South Korea make most 300 mm wafers used for advanced chips. China makes most polysilicon, almost all of it for solar cells.



Specimen 02 · Silicon and Wafers

A silicon ingot, one crystal pulled whole from a melt, with two wafers sawn from it. It is 2.1 m long, 300 mm across and about 265 kg, taller than a door and about as heavy as three grown men. It hangs from a neck about 10 mm wide. Drawn to scale.

Parts, as numbered on the drawing

1. Seed: lowered to the melt and drawn slowly back up
2. Neck: keeps the seed's first flaws out of the ingot
3. Ingot: grown by Shin-Etsu Handotai (JP), SUMCO (JP), GlobalWafers (TW), SK Siltron (KR) and Siltronic (DE)
4. Wafers: sawn from the ingot, then lapped, etched and polished

In plain terms

Every chip is built on a wafer, a thin round slice of silicon about 30 centimeters across that carries hundreds of chips at once. Making one starts with quartz rock, which is silicon bound to oxygen[1]. The silicon is refined until metal impurities are no more than a few parts per billion[2]. It is then melted, and a small seed crystal is lowered to the

1 Mineral Commodity Summaries 2026: Silicon. U.S. Geological Survey, 5 February 2026. pubs.usgs.gov
2 SUMCO, production processes page. SUMCO, accessed 9 September 2026. sumcosi.com

surface and drawn slowly back up. The melt freezes onto the seed as one long cylinder, a single crystal with its atoms lined up in one unbroken pattern from end to end. Saws cut the cylinder into discs, and each disc is polished to a mirror finish and made extremely flat across its whole face[2]. In 2025 the top five wafer makers served around 75 percent of the world market, and none of them is American or Chinese[3].

In short

Japanese companies held 53 percent of the world's silicon wafer sales in 2023, by a Fuji Keizai survey that Japan's trade ministry reproduced[4]. Firms based in Taiwan, South Korea and Europe held most of the rest[4]. China's largest maker of 300 mm wafers, Eswin, says it ranked sixth in the world with about 7.7 percent of global 300 mm shipments in the first half of 2026[5]. Chips also need electronic-grade polysilicon, the raw material for wafers[6]. The German wafer maker Siltronic named five major suppliers of it for 2025, and none is Chinese[6].

GlobalWafers of Taiwan opened a $3.5 billion plant in Texas in May 2025, the first fully integrated plant for advanced silicon wafers built in the United States in over twenty years[7]. US Presidential Proclamation 11052 sets a minimum import price of $100 per kilogram for polysilicon ingots and wafers from 4 December 2026[8].

The world market for silicon wafers was $11.4 billion in 2025, on 12,973 million square inches shipped[9]. That is a small fraction of Nvidia's data center revenue, which reached $193.7 billion in the fiscal year to 25 January 2026[10].

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Chokepoint card

Silicon and Wafers

Concentration

High

Four to six firms could supply this stage today, and the top two take most of the revenue.

Substitutability

Moderate

Several firms already make 300 mm wafers that fabs have approved, but customers have to test and approve a new wafer plant before they buy from it [11].

Price or market size

$11.4 billion

world wafer market in 2025, on 12,973 million square inches shipped [9]

Who leads	JP	**Shin-Etsu Handotai**	Its parent's electronics materials segment, wafers included, had ¥1,015.7 billion of sales in the year to March 2026 [12]
	JP	**SUMCO**	¥409,670 million of net sales in 2025 [13]
	TW	**GlobalWafers**	NT$60.6 billion revenue in 2025 [14]
	KR	**SK Siltron**	One of the top five wafer makers by revenue in the first three quarters of 2025 [3]
	DE	**Siltronic**	€1,346.7 million revenue in 2025 [15]
Where it is made	JP	**Japan**	Shin-Etsu and SUMCO crystal growth and polishing
	TW	**Taiwan**	GlobalWafers headquarters and plants
	DE	**Germany**	Siltronic plants in Burghausen and Freiberg, and Wacker semiconductor-grade polysilicon
	KR	**South Korea**	SK Siltron
	US	**United States**	Hemlock polysilicon in Michigan, and GlobalWafers plants in Sherman, Texas and St Peters, Missouri

Why substitution is possible

Losing one of the top five wafer makers leaves four, and a new wafer plant can still be built [3]. Customer approval of a new plant is called qualification [11]. Siltronic completed key customer qualifications at its new Singapore fab in July 2025 [11].

Where China stands

China made about 93 percent of the world's polysilicon in 2023, and an estimated 98 percent of China's output was solar-grade [16]. Eswin, mainland China's largest maker of 300 mm wafers, says it had about 7.7 percent of world 300 mm shipments in the first half of 2026 [5].

Where the US stands

None of the five largest wafer makers in 2025 was American [3]. GlobalWafers of Taiwan won CHIPS awards of up to $406 million in December 2024 for wafer plants in Texas and Missouri [17].

3 Investor Presentation: Foundation of Digital Life. Siltronic, April 2026, p. 6. siltronic.com

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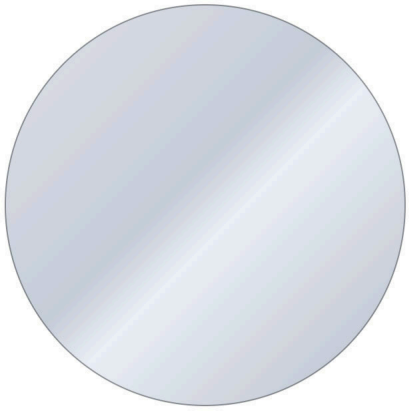
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01

How silicon wafers are made

Figure 02.1 · Explainer, 20 seconds

From sand to wafer



POLISHED TO A MIRROR FINISH

A few parts per billion
METAL IMPURITIES

1,420 °C
MELTING POINT, HOTTER THAN LAVA

30 cm
ACROSS ONE FINISHED WAFER

1. **Refine.** Silicon starts as quartz rock. It is refined into rods so pure that metal impurities are no more than a few parts per billion.
2. **Grow.** The rods are melted. A seed crystal touches the melt and is drawn slowly up, turning, and the silicon freezes onto it as one long crystal.
3. **Slice.** A wire saw slices the crystal, called an ingot, into thin discs.
4. **Polish.** Each disc is ground, etched and polished to a mirror finish, flat right out to the edge.

Note: Simplified. The ingot and the machines are not drawn to scale.

An arc furnace melts the quartz rock with carbon, and the carbon takes the oxygen away as gas, leaving rough silicon metal. Refiners turn the metal into a gas, trichlorosilane, and distill it, because the impurities boil at other temperatures and separate out.

In the next step, the Siemens process, the clean gas flows over silicon filaments heated inside a bell jar. On the hot surface the gas breaks apart and its silicon settles onto the filaments. Over days the filaments thicken into gray rods of polysilicon. Everything downstream is made from those rods. Polysilicon comes in two ultra-high-purity grades, one for semiconductors and one for solar cells, and chips use the semiconductor grade[1].

A grower melts the polysilicon at about 1,420 degrees Celsius, hotter than lava, and pulls a single crystal out of it by the Czochralski method[2]. The grower lowers a small seed crystal on the end of a rod to the surface of the melt and draws it back up, slowly turning. Silicon freezes onto the seed in the seed's own atomic pattern, and a cylinder, the ingot, grows beneath it. The pull starts with a thin neck so that flaws from the

1 Mineral Commodity Summaries 2026: Silicon. U.S. Geological Survey, 5 February 2026. pubs.usgs.gov
2 SUMCO, production processes page. SUMCO, accessed 9 September 2026. sumcosi.com

seed's first contact with the melt stay in the neck and never reach the ingot. Boron or phosphorus is stirred into the melt, a step called doping. The added element sets how readily the wafer conducts and whether negative or positive charges carry its current.

A wire saw, a web of fine wire running through abrasive, slices the ingot into discs. The maker then laps each disc, grinding both faces flat between rotating plates. It etches off the damaged surface with chemicals, then polishes the disc to a mirror with colloidal silica, a slurry of silica particles far too small to see. Almost every step throws material away.

Epitaxial wafers get one more layer of near-perfect silicon, grown on top at about 1,200 degrees Celsius[2]. That is still hot enough to glow orange.

02

Types of silicon wafers

Polished versus epitaxial

A polished wafer is a doped single crystal polished to a mirror finish[2]. An epitaxial wafer is a polished wafer with a thin layer of single-crystal silicon grown on top[2]. The transistors are built in that layer, so flaws in it can make them leak current. SUMCO says an epitaxial wafer suits customers who need a near-perfect crystal or layers that differ in electrical resistance[2]. SEMI, the industry association, credits the 2025 rise in shipments to AI-driven demand for advanced epitaxial wafers for logic chips and polished wafers for high-bandwidth memory[9].

Silicon on insulator

In a silicon-on-insulator (SOI) wafer, a buried layer of oxide, which is glass, separates a thin working layer of silicon from the bulk beneath. Soitec says nearly all SOI wafers sold worldwide are made with its Smart Cut process[18]. Soitec's SOI product lines include RF-SOI for radio chips and fully depleted SOI (FD-SOI)[18]. Its markets run from mobile communications to cloud AI[18]. But for logic chips, SEMI credited the AI-driven growth in 2025 to a different kind of wafer, the advanced epitaxial wafer[9].

03

Who makes silicon wafers and polysilicon

The Center for Security and Emerging Technology (CSET) at Georgetown University named five key wafer firms in a January 2021 brief[19]. They were Shin-Etsu and SUMCO of Japan, GlobalWafers of Taiwan, Siltronic of Germany and SK Siltron of South Korea[19]. CSET also listed nine Chinese-owned producers, among them Zhonghuan, Simgui, ZingSEMI and Okmetic, a Finnish firm bought by a Chinese state investment

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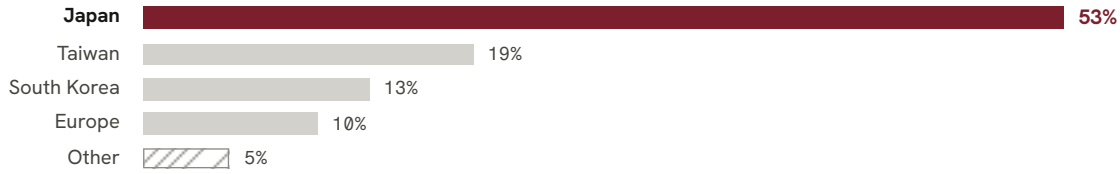
fund[19]. It said Chinese wafer producers made little revenue compared with the five key firms[20].

China's 300 mm wafer industry has grown since 2018. As of 2018, five Chinese firms were making or planning 300 mm wafers[20]. In its October 2025 listing prospectus, Eswin, a Chinese wafer maker, counts seven Chinese firms, itself included, making 300 mm wafers at scale[21]. Their combined capacity was 2.94 million wafers a month at the end of 2024[21].

Figure 02.2

Companies based in Japan held 53 percent of 2023 wafer sales

Silicon wafer sales by supplier headquarters, 2023. The source shows no separate share for China or the United States



Source: METI, December 2025, from Fuji Keizai data (2023). Japan's and Europe's shares are printed, and the others are read off METI's chart [4]

Japanese companies held 53 percent of the world's silicon wafer sales in 2023[4]. The shares come from a survey of 2023 sales by Fuji Keizai, a Japanese research firm[4]. Japan's Ministry of Economy, Trade and Industry reproduced the figures in a December 2025 strategy deck[4]. The ministry's chart of the survey prints only Japan's 53 percent, so the other shares are measured from its bars[4]. An earlier page of the ministry's deck prints Europe's share in the same Fuji Keizai survey as 10 percent[22]. The ministry's figures give no separate share to Chinese or US companies[4].

Siltronic's April 2026 investor presentation says the top five wafer makers serve around 75 percent of the market, based on revenue in the first three quarters of 2025[3]. It names the five as Shin-Etsu, SUMCO, GlobalWafers, SK Siltron and Siltronic itself[3]. Siltronic describes itself as the only Western-based maker among the five[3].

The wafer makers' 2025 results show strong AI demand and weak demand for chips made on older, mature nodes.

- **Shin-Etsu.** Shin-Etsu Chemical's electronics materials segment, which includes its wafer business, sold ¥1,015.7 billion in the year to March 2026, up 9 percent[12]. AI-related demand stayed strong and demand in other sectors began to rise, lifting sales of silicon wafers, photoresist and photomask blanks[12]. The segment also sells rare earth magnets and other materials, and the results release does not break out

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wafer sales[23]. So the figure cannot be compared with the wafer revenue of the other firms listed.

- **SUMCO.** Sold ¥409,670 million[24]. Its operating profit fell from ¥36,924 million to ¥1,342 million, and it ended the year with a net loss of ¥11,751 million[24]. In the same year it added leading-edge 300 mm capacity and reorganized production of 200 mm and smaller wafers, where demand stayed weak[13].
- **Siltronic.** Lost €77.9 million on €1,346.7 million of sales while investing €369.1 million, with its new Singapore fab as the focus[15].
- **GlobalWafers.** NT$60.6 billion of revenue, down 3.24 percent in local currency[14].

Worldwide, the wafer area shipped rose 5.8 percent in 2025, but revenue fell 1.2 percent[9]. SEMI attributes the softer revenue mostly to traditional chip uses outside AI, where demand and pricing had yet to improve[9].

Polysilicon is the raw material for both wafers and solar cells, and almost all of China's is made for solar cells. In 2023 China made 1.50 million metric tons of polysilicon, about 93 percent of world output[16]. Solar-grade material was an estimated 98 percent of that output and electronic-grade 2 percent[16]. The US Commerce Department found in 2026 that semiconductor-grade polysilicon makes up only 2.4 percent of global production, close to the electronic-grade share of China's output[8]. These figures do not show how much of China's electronic-grade polysilicon ends up in chips.

Figure 02.3

An estimated 98 percent of China's 2023 polysilicon was solar-grade

China's polysilicon output by grade, 2023. China made 1.50 million metric tons that year, about 93 percent of world output



Source: USGS Minerals Yearbook, China, 2023 [16]

Siltronic put the 2025 market for electronic-grade silicon at $1.4 billion, with five major suppliers, none of them Chinese: Wacker, Hemlock, OCI, Tokuyama and Mitsubishi[6].

GlobalWafers of Taiwan has expanded its wafer making in the United States. It opened a $3.5 billion plant in Sherman, Texas in May 2025, the first fully integrated plant for advanced silicon wafers built in the United States in over twenty years[7]. The same day it announced another $4 billion for phases three and four in Sherman[7]. The US government had awarded it up to $406 million under the CHIPS incentives program for wafer plants in Texas and Missouri[17].

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US import rules issued in 2026 on polysilicon products also cover the single-crystal wafers that chips are built on. US Presidential Proclamation 11052, signed 6 August 2026, sets a minimum import price of $100 per kilogram for polysilicon ingots and wafers from 4 December 2026[8]. From the same date it adds an extra 15 percent duty on imported ingots and other listed products made from polysilicon[8]. A September 2026 US Commerce Department rule meant to stop stockpiling before the duties start lists tariff code 3818.00.0040 among the products it covers[25]. That code covers round, doped, single-crystal silicon wafers[26].

The US Commerce Department may approve plans to build US polysilicon, ingot, wafer or solar cell plants whose construction starts by 20 January 2029[8]. Firms with approved plans can import production equipment, and polysilicon, ingots, wafers and solar cells in amounts the department sets to match their new investment, without the new duties[8].

04

Why 300 mm wafers are a chokepoint

Almost all fab capacity for advanced chips runs on 300 mm wafers[20]. In 2025 the top five wafer makers served around 75 percent of the world market, and none of them is American or Chinese[3]. Losing one of the top five would leave four, and a new wafer plant can still be built[3]. But customers have to qualify a new plant before they buy from it, and Siltronic completed key customer qualifications at its new Singapore fab in July 2025[11].

China makes most of the world's polysilicon, but as of 2019 Chinese firms held under 5 percent of the wafer market and under 1 percent of the 300 mm market[19]. As of 2018, Chinese producers made 12 percent of the wafers for Chinese-headquartered 300 mm fabs, against 18 percent for 100 to 200 mm fabs[27].

Eswin, mainland China's largest maker of 300 mm wafers, says it ranked sixth in the world with about 7.7 percent of global 300 mm shipments in the first half of 2026[5]. Its October 2025 Shanghai listing prospectus, drawing on SEMI data, estimates that the five largest makers shipped 79.08 percent of 300 mm wafers in 2024[28]. It puts Eswin at 6.01 percent and all other makers at 14.91 percent[28].

Making the smoothest, high-purity 300 mm wafers takes considerable know-how that is hard to write down[20].

3 Investor Presentation: Foundation of Digital Life. Siltronic, April 2026, p. 6. siltronic.com

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In SEMI figures from November 2020 cited by CSET, 99.7 percent of world fab capacity at 45 nm and below used 300 mm wafers[20]. SEMI estimated that 300 mm wafers made up about 78.8 percent of all wafer area shipped worldwide in 2025, according to Eswin's 2025 annual report summary[29].

The last increase in wafer size, to 300 mm, happened in 2002[30]. Wafers are still produced in diameters of up to 300 mm[9]. Bigger wafers cut the cost of deposition, etching and other steps, because those steps have to be run fewer times for the same number of chips[30]. So printing, or photolithography, takes a bigger share of the processing cost[30]. A 2013 University of Texas lecture cited by CSET put photolithography at half the cost of processing a 300 mm wafer, against a quarter for a 150 mm one[30].

Large GPU dies and the newest processes need better wafers. Some GPU dies fill the whole area a lithography tool can print in one exposure. One fatal defect on such a die ruins far more silicon than one on a small chip. Extreme-ultraviolet printing also stays in focus over only a tiny range of heights (see Lithography), so the disc has to be flat at every point. SEMI's wafer group says the move to sub-3nm processes is raising requirements for wafer quality and consistency[9].

Four questions to answer from memory

1. What is a wafer? `see How it works`
2. Which five companies make most of the world's silicon wafers? `see Who makes it`
3. Why can a fab not switch wafer suppliers quickly? `see The chokepoint`
4. Why does making most of the world's polysilicon give China little hold over chip wafers? `see The chokepoint`

Four things to remember

- Every chip starts on a wafer, a polished disc of pure silicon about 30 centimeters across.
- The top five wafer makers, in Japan, Taiwan, Germany and South Korea, served around 75 percent of the market in 2025.
- Customers have to test and approve a new wafer plant before they buy from it.
- China makes most of the world's polysilicon, but none of the five major suppliers of electronic-grade silicon, the grade chips need, is Chinese.

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A company filings, governments, standards bodies, universities and research institutions; B SemiAnalysis, Reuters, the Financial Times and Nikkei Asia.

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